

Stack Overflow Developer Survey Analysis

Tools

Python, Pandas, Jupyter Notebook

Description

Comprehensive analysis of Stack Overflow Developer Survey data examining developer demographics, compensation trends, programming languages, learning paths, and remote work patterns using descriptive statistics and data aggregation techniques.

My Tasks

- Large dataset exploration and data quality assessment
 - Descriptive statistics and central tendency measures calculation
 - Multi-criteria data filtering and cohort analysis
 - Compensation analysis across geographic and demographic segments
 - Popular technology and industry trend identification
 - Missing value handling and data completeness evaluation
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Result

- Processed 10,000+ survey responses identifying complete response rates and data integrity
 - Calculated mean, median, and mode for developer experience across the dataset
 - Quantified Python adoption rate and remote work prevalence among respondents
 - Built compensation analysis by country for Python developers with mean/median metrics
 - Identified education levels of top 5 highest-earning professionals
 - Analyzed Python popularity across age demographics and high-earning remote workers' industries
 - Delivered actionable insights on developer market trends and compensation patterns
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